

7.1.36. Let $V = \{1, 2, 3\}$. Student C tries to define a function $S : V \rightarrow V$ as follows: For each $x \in V$, $S(x)$ is the integer y in V so that $(xy) \bmod 4 = 1$. Student D claims that S is not well defined. Who is right: student C or student D? Justify your answers.

$$S(x) : V \rightarrow V$$

Definition of a well-defined function $f : X \rightarrow Y$:

- (1) Elements of X are related to Y .
- (2) Elements of X can only relate to one element of Y .

Let $V = \{1, 2, 3\}$

$$V \times V = \{ \\ (1, 1), (1, 2), (1, 3), \\ (2, 1), (2, 2), (2, 3), \\ (3, 1), (3, 2), (3, 3), \\ \}$$

Hence, there're 9 possible (x, y) pairs.

Suppose $(xy) \bmod 4 = 1$

Then:

$$\begin{aligned} x = 1, y = 1 : (1 * 1) \bmod 4 &= 1 \\ x = 1, y = 2 : (1 * 2) \bmod 4 &= 2 \\ x = 1, y = 3 : (1 * 3) \bmod 4 &= 3 \\ x = 2, y = 1 : (2 * 1) \bmod 4 &= 2 \\ x = 2, y = 2 : (2 * 2) \bmod 4 &= 0 \\ x = 2, y = 3 : (2 * 3) \bmod 4 &= 2 \\ x = 3, y = 1 : (3 * 1) \bmod 4 &= 3 \\ x = 3, y = 2 : (3 * 2) \bmod 4 &= 2 \\ x = 3, y = 3 : (3 * 3) \bmod 4 &= 1 \end{aligned}$$

There are only two cases where $(xy) \bmod 4 = 1$, for $x = 1, y = 1$ and $x = 3, y = 3$.

Student C claims that for each $x \in V$, $S(x)$ is the integer y so that $(xy) \bmod 4 = 1$.

For $x = 1$, there is $y = 1$:

$$(1 * 1) \bmod 4 = 1$$

For $x = 3$, there is $y = 3$:

$$(3 * 3) \bmod 4 = 1$$

But for $x = 2$, there is no $y \in V$, such that $(2y) \bmod 4 = 1$:

$$(2 * 1) \bmod 4 = 2$$

Therefore, the function is not well-defined.

Solution: Student D is right: The function is not well-defined.